

Introduction to Macroeconomics

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Problem Set 1 2026

1 Introduction

Exercise 1 (Exam August 2013)

Which of the following statements are correct ?

- (I) GDP and the wealth of a person are stock variables.
 - (II) According to monetarists monetary policy is inefficient if it is not anticipated.
 - (III) A correlation between two variables, say x and y , always implies that there is a causal relation between those two. The goal of an economist is to put forward hypotheses about this causal link and to test them using available data.
 - (IV) In an economic model endogenous variables are explained by exogenous variables.
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- (a) Only IV is correct.
 - (b) Only I is false.
 - (c) II and III are correct and the rest is false.
 - (d) II and IV are correct and the rest is false.

2 Gross domestic product

Exercise 2 (Exam June 2014)

Assume that countries A and B can only trade with each other. We have the following information :

Country A		Country B	
GDP at market prices	110	Capital depreciation	15
Government expenditure	25	Subsidies	10
Gross investment	20	Indirect taxes	5
Net exports	15	National income	80
Factor income paid to foreigners	10		
Factor income received from foreigners	15		
Indirect taxes	10		

We also know that in Country A, capital depreciation is equal to subsidies minus indirect taxes. Amongst the following propositions, which one is correct ?

- (a) Country A's NNP at factor prices equals 115 and country B's GDP at market prices equals 100.
- (b) Country B's GDP at factor prices equals 120 and country A's national income equals 110.
- (c) Country A's GNP at market prices equals 105 and country B's GDP at market prices equals 95.
- (d) Country B's GDP at market prices equals 95 and country A's NNP at factor prices equals 115.

Exercise 3

The Prime Minister of a neighbouring country claims during the re-election campaign that the (nominal) GDP has increased by 12% during his/her last four-year term in office. We know that during this four-year period, the GDP deflator has increased from 100 to 108.

- Write down the relationship between the nominal GDP and the real GDP and derive mathematically the relationship between the percentage change in nominal GDP and the percentage change in real GDP.
- Calculate the percentage change in real GDP during the Prime Minister's term in office

Exercise 4 (problem 5, chapter 23 Mankiw-Taylor)

Below is some data on a country which produces only milk and honey. Prices are given in €/litre.

Year	Price of milk	Quantity of milk (litres)	Price of honey	Quantity of honey (litres)
2003	1	100	2	50
2004	1	200	2	100
2005	2	200	4	100

1. Compute the nominal GDP, real GDP and the GDP deflator for each year, using 2003 as the base year.
2. Compute the percentage change in nominal GDP, real GDP and the GDP deflator in 2004 and 2005 with respect to the preceding year. For each year, identify the variables that do not change. Explain in words why your answer makes sense.
3. Did the economic well-being increase more in 2004 or 2005 ? Explain.

Exercise 5

The economy of Superland is made up of three sectors : Agriculture, Manufacturing and Services. Here is the interindustrial transactions table :

	Agriculture	Industry	Services	C	I	X	M	Total production
Agriculture	1	A	1	6	1	4	3	13
Industry	2	10	4	10	5	8	5	34
Services	1	4	1	6	4	2	3	15
Compensation of employees	5	7	4					
Capital revenue	3	6	3					
Profits	1	4	2					
Total production	13	34	15					

1. How would you interpret this table ? What does the letter A represent ? Calculate its value.
2. For this simple economy, calculate the GDP according to the three approaches (production, income, expenditure) using the data from the table.

Exercise 6 (Question 11, chapter 23, Mankiw-Taylor, 3rd francophone edition, 2013)

Participation of women in the active workforce has considerably increased over the last thirty years in a number of European and North-American countries.

1. How do you think this phenomenon could have influenced the value of GDP in these countries ?

2. Imagine now that there is a welfare indicator which takes into account household work and leisure time. How do you think this indicator would have evolved in response to the above phenomenon as compared to the evolution in the GDP ?
3. Can you think of other aspects of economic well-being that could have been affected by the increase of female participation in the active work-force ? Would it be easy to develop a measure of economic well-being which could encompass all these aspects ?

3 Additional exercises (will not be discussed in the seminar)

Exercise 7

The Gross domestic product (GDP) of a country :

1. Only includes the final goods and services.
2. Only includes the goods and services produced by national firms.
3. Measures the market value of the totality of the national output.
4. Only includes the final output of that national territory.

Which of the above propositions are correct ?

- (a) I, III and IV
- (b) II and III
- (c) All are correct
- (d) I and IV are correct

Exercise 8 (Exam June 2013)

Assume that countries A and B trade only with each other.

Country A		Country B	
Consumption	60	Net factor income from abroad	5
Investment	30	National Income	40
Government spending	40		
Exports	10		
Imports	15		
Subsidies	10		
Depreciation	10		
Indirect taxes	5		

Which of the following statements is correct ?

- (a) Country A's GDP is lower than its GNP
- (b) GDP calculated using the expenditure approach equals GDP plus income received by national firms from abroad.
- (c) Country A's national income is higher than that of country B
- (d) Country B's GNP at factor prices is lower than 40.